

KANNAN APPIYA SANTHARAM

Senior Lead Software Engineer (Lead Frontend Engineer) | React · Node.js · AI-Native Engineering

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Nationality: Indian | Notice Period: 60 Days

PROFESSIONAL SUMMARY

Senior Lead Software Engineer with 10.5+ years of experience turning AI-native engineering into measurable business impact: architected an AI test-authoring platform and LLM-orchestrated agent workflows, backed by custom Claude Code skills and Model Context Protocol (MCP) servers with strict write-scope guardrails, cutting manual test effort and speeding releases. Delivered a 96% build-time reduction reclaiming hundreds of engineering hours monthly and eliminated 30,000+ lines of dead code, without disrupting a platform serving 4,000+ MSP and IT enterprise customers. Proven record leading solo platform migrations and mentoring engineering teams. Seeking a Senior Lead Software Engineer / Engineering Manager role in India.

CAREER HIGHLIGHTS

- Architected an AI test-authoring platform from zero, orchestrating LLM agents over HTTP Streamable Web Streams (fetch + ReadableStream) to auto-generate, execute, and self-heal end-to-end test suites, while engineering custom Claude Code skills and Model Context Protocol (MCP) servers with strict write-scope guardrails.
- Reduced cold-start build compilation from 2 minutes to 5 seconds, a 96% improvement, through a solo Webpack 5 to Rspack migration completed in 3 weeks across 12 monorepo packages.
- Led an engineering squad of frontend developers at SuperOps, driving sprint execution, architecture reviews, and cross-team delivery supporting 4,000+ MSP and IT enterprise customers.
- Mentored 6 to 7 junior and onboarding software engineers across SuperOps and Freshworks on React best practices, REST API development, and UI architecture to accelerate team ramp-up.

TECHNICAL SKILLS

AI & Agentic Tooling: Claude Code, Model Context Protocol (MCP) Servers, Cursor IDE, LLM Orchestration, HTTP Streamable (ReadableStream), Agent Guardrails & Write-Scope Design

Frontend Core: React, TypeScript, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, SASS, Responsive Web Design, Core Web Vitals

State, APIs & Streaming: GraphQL, Apollo Client, Zustand, REST API Design, Custom Cached Query Hooks, Server-Sent Events (SSE), HTTP Streamable Web Streams

Architecture: Monorepos, Micro-Frontends, Design Systems, Role-Based Access Control (RBAC), Module Federation

Build & Performance: Rspack, Webpack 5, Code Splitting, Knip Static Analysis, Bundle Optimisation

Testing, Cloud & DevOps: Playwright End-to-End Suite (232 specs), Node.js, Jenkins CI/CD Pipelines, Docker, AWS (EC2, S3, Route 53), Amazon Bedrock AgentCore

Leadership & Process: Engineering Squad Leadership, Developer Mentorship (6 to 7 engineers), Technical Knowledge Sharing, Agile Methodology, Git & GitHub Flow, Architecture & Code Reviews, Cross-Team Collaboration

PROFESSIONAL EXPERIENCE

Senior Lead Software Engineer

Jul 2022 – Present

SuperOps — Chennai, India

Progression: Senior Software Engineer → Lead Software Engineer → Senior Lead Software Engineer

- Led an engineering squad of frontend developers, driving sprint execution, architecture reviews, and cross-team collaboration across product, design, and QA to deliver enterprise features supporting 4,000+ MSP and IT enterprise customers.
- Mentored junior and mid-level engineers on React best practices and monorepo code quality, while standardising a reusable UI SDK framework adopted across product teams.
- Architected an AI test-authoring platform from zero using React and Node.js, orchestrating LLM agents over HTTP Streamable Web Streams (fetch + ReadableStream) to auto-generate, run, and self-heal

end-to-end test suites.

- Authored custom production Claude Code skills and engineered Model Context Protocol (MCP) servers, establishing strict multi-tier write-scope guardrails to secure autonomous agent workflows across the codebase.
- Built a headless design system from scratch on Base UI, and authored a custom Claude Code skill that pulls design context from a Figma node link through an MCP server to generate production components automatically, cutting page build time from 2 days of AI-assisted engineering to a few hours.
- Executed a solo Webpack 5 to Rspack migration in 3 weeks, reducing cold-start build compilation from 2 minutes to 5 seconds across 12 packages.
- Improved application load times through React lazy loading, route-level code splitting and per-package chunking strategy across the monorepo.
- Eliminated 30,000+ lines of dead code and unused dependencies across the monorepo using Knip static analysis.
- Maintained platform health with a 232-spec Playwright regression suite in Jenkins CI.
- Built role-based access control (RBAC) gates and high-frequency polling dashboards using custom cached Apollo query hooks, balancing data freshness against network load.

Senior Software Engineer (Customer-Facing Engineering)

Jun 2018 – Jul 2022

Freshworks — Chennai, India

Progression: Onboarding Engineer → Senior Software Engineer

- Mentored 6 to 7 junior and onboarding software engineers on React, REST API development, and UI architecture, conducting regular code and design reviews to accelerate ramp-up time.
- Worked directly with customers alongside Support, Customer Success and Sales teams to understand use cases and requirements, then partnered with Product Design to deliver the resulting UI.
- Built UI web pages and REST API services across Freshworks products using React, JavaScript, Node.js, HTML and SASS.
- Built a Customer 360 dashboard consolidating all of a customer's information into a single operational view.
- Integrated enterprise SaaS platforms including Jira, Salesforce, Zendesk and ServiceNow with Freshworks products through REST API services.
- Implemented multiple Freshworks add-ons and supporting REST API services to solve customer-specific operational use cases.
- Created UI wireframes and design mockups in Balsamiq, and led code and UI design reviews.

Software Developer

Mar 2016 – Jun 2018

Niche Video Media, LLC / Infigenic, LLC (acquired by Freshworks) — Chennai & Bengaluru, India

Progression: Web Application Developer & Designer → Software Developer

- Designed and implemented web pages, landing pages, and plugins for a commercial video-hosting platform.
- Built a video player customisation feature from scratch with configurable player buttons, annotations and call-to-action overlays.
- Implemented an admin dashboard that dynamically generated Stripe pricing plans based on feature selection, storage, and bandwidth tiers.
- Built high-converting company and product landing pages supporting top-of-funnel demand.
- Built an integration connecting Freshservice with DocuSign using REST API services, joining Freshworks with the team on acquisition.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science

2011 – 2015

K.L.N. College of Information Technology, Tamil Nadu, India

LANGUAGES

English: Fluent (Professional Working Proficiency) | **Tamil:** Native | **Hindi:** Intermediate